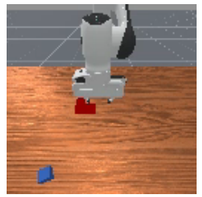
4D MAE / B5

PullCube configuration

(a) Policy architecture



Current RGB

ResNet18
RGB stem + resampler

16 tokens

Robot state

State MLP
+ resampler

16 tokens

Point history C

Tokenizer Tθ
+ point resampler

16 tokens

Learned queries
64 action + 16 future

HPT
128 × 256
16 blocks
8 heads

Concat. + position

64 action

Noisy actions + τ

Action flow head
6 blocks · width 128

Action chunk
32 × 8
Execute 8; replan

16 future-context tokens (K, V)

Point query Q
Current XYZ

Tθ
Point token

q

Trajectory decoder
q attends to K, V

Noisy tracks + flags

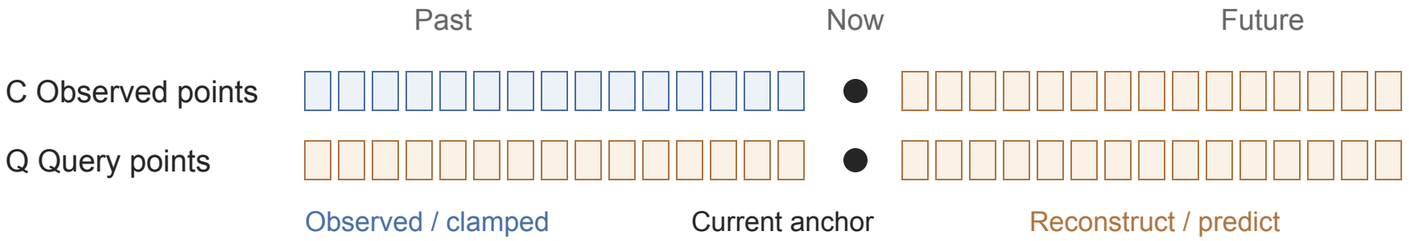
Flow time τ

Q completion
50 flow steps
15 past + 15 future

Tθ: shared weights

Control uses the action branch. Trajectory completion supplies auxiliary training supervision.

(b) Trajectory supervision



$$L = L_{action} + (L_{C,future} + L_{Q,past} + L_{Q,future}) / 3$$

Flow-matching losses on valid targets.

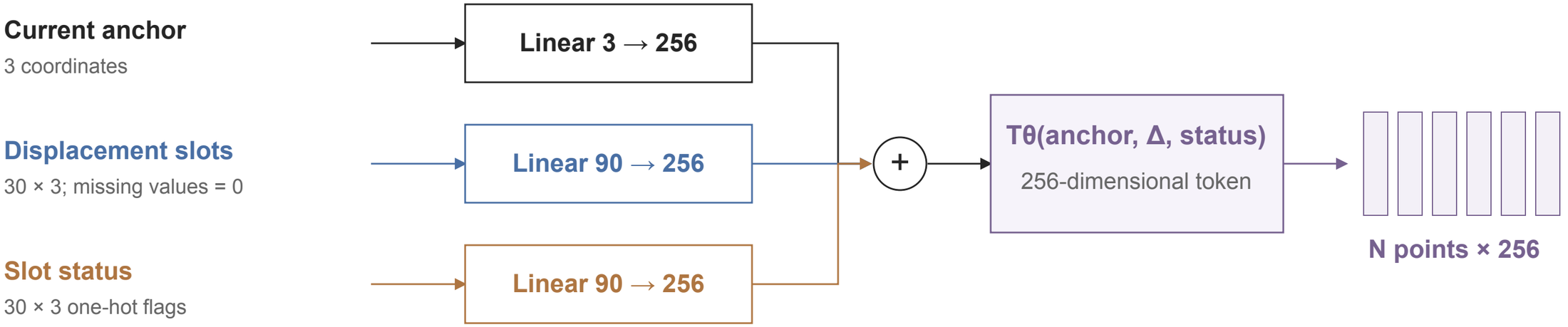
C: same decoder; observed past stays fixed.

64 C points · 64 Q points · 16 history frames · privileged simulator history in PullCube

All trajectory sketches are schematic.

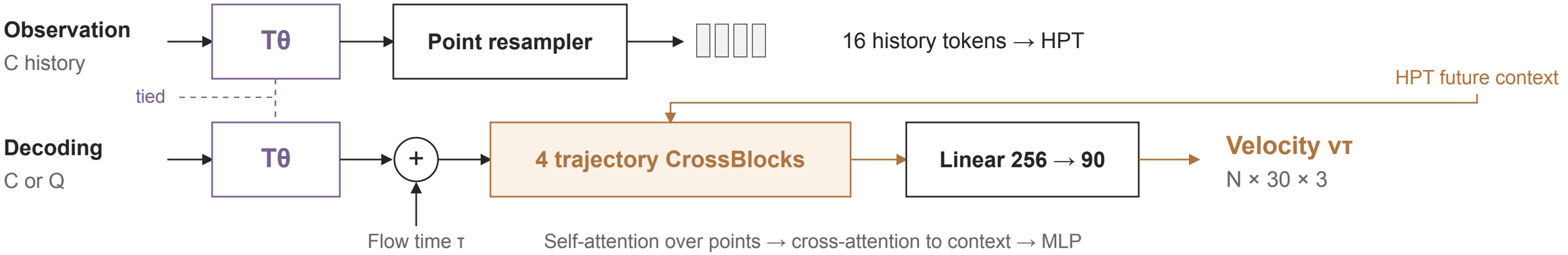
B5: shared trajectory tokenizer

(a) One trajectory → one point token



Status: missing / observed / noisy. All 30 time slots are packed into each point token.

(b) The same Tθ in both paths



Sampling: 50 Euler steps. Observed C past is clamped at every step.

τ is generative time; the 30 output slots are physical time. Add the current anchor to recover absolute XYZ.